TriOPS: QR Production Process and Salary Management System and Workforce Analytics in Tagum Trio Lumber Corporation

A Capstone Project Presented to

Faculty of the Institute of Computing

Davao del Norte State College

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Chapter I
INTRODUCTION

A. Background of the Study

Manufacturing industries, particularly those involved in wood processing and plywood production, play a significant role in economic growth and industrial development. These industries align with the United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation, and Infrastructure), which promotes sustainable industrialization and the adoption of innovative technologies to improve efficiency and productivity [1]. In plywood manufacturing, the transformation of raw materials into finished products involves multiple interconnected processes such as log receiving, veneer cutting, drying, pressing, and finishing. However, many manufacturing companies still rely on manual monitoring systems, resulting in inefficiencies, delays, and limited real-time visibility of operations. [2], the transition toward digital and automated systems is essential for improving manufacturing performance under Industry 4.0. It emphasized the connected and smart systems enable real-time data collection, which enhances operational efficiency and decision-making in industrial environments.[3]

The Department of Science and Technology (DOST), through the Harmonized National Research and Development Agenda (HNRDA), the role of science, technology, and innovation in enhancing industrial productivity and ensuring that research outputs generate economic and social benefits for the country[4], Real-time production monitoring systems have been widely recognized for their ability to improve manufacturing performance. Studies show that such systems help identify bottlenecks, reduce downtime, and enhance productivity by providing immediate feedback on production activities.[5] Furthermore, integrating labor monitoring with production systems improves workforce management and accountability. Performance-based compensation systems can increase worker productivity when supported by accurate tracking systems. However, manual payroll computation often leads to errors and

delays, highlighting the need for automated payroll systems integrated with production data.[6]

At the institutional level, the proposed study aligns with the Davao del Norte State College (DNSC) Research, Development, and Extension (RDE) Agenda, which focuses on sustainable development, innovation, and economic productivity within the region. The DNSC framework emphasizes that research initiatives should contribute to environmental conservation, community resilience, and economic productivity, in line with the Sustainable Development Goals [7]. This is supported by local studies in the Philippines which highlight that digital transformation and information systems significantly improve operational efficiency and decision-making in small and medium enterprises (SMEs). These findings reinforce the relevance of implementing real-time monitoring and payroll systems in manufacturing settings such as plywood production.

B. Company Overview

Tagum Trio Lumber Corporation is a prominent plywood manufacturing enterprise established in June 2025 located at Brgy Pinagkaisa Lapaz Carmen Davao del Norte and is currently under the capable management of Mr. Jiang Kuanrong. As the company's head, Mr. Jiang oversees the overall operations and spearheads the strategic direction of the business, ensuring sustainable growth and operational excellence. As a developing manufacturing firm, the company is primarily engaged in the production of high-grade plywood products designed for various applications, including residential and commercial construction, furniture making, cabinetry, and other industrial uses. With a substantial workforce of approximately 300 employees, the company plays a vital role in the local economy by providing significant employment opportunities and actively supporting economic activity within the region.

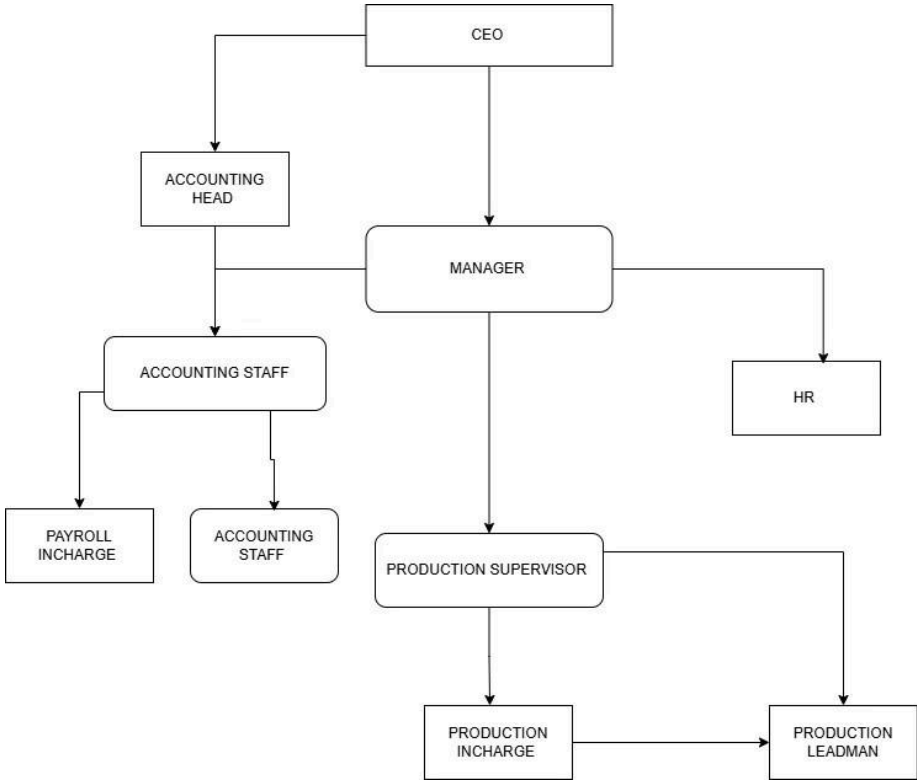

Figure 1. Organizational Structure

C. Problems Encountered

The Production Head of Tagum Trio Lumber Corporation is responsible for overseeing approximately 300 employees within the production department alone. Currently, the manual management of essential tasks—such as maintaining employee records, processing salaries, tracking leave requests, handling excused absences, and monitoring half-day schedules—has become a significant challenge. This outdated method places a heavy administrative burden on the team and creates unnecessary complexities. Consequently, it leads to delays, reduces overall efficiency, and significantly slows down daily operational processes. The key problems encountered by Tagum Trio Lumber Corporation are as follows:

- Inefficient data management due to reliance on paper records and manual entry.
- Time-consuming preparation of production and payroll reports.
- Lack of material traceability from raw state to finished goods.
- Delayed decision making due to lack of real-time production data.
- Unclear performance evaluation with no standard way to compare production rates and workloads.
- Difficulties in tracking staff deployment and transfers between departments.

D. Existing Business Process

Record products process (done)

Figure . 2

Salary Process

The figure below illustrates the current process of distributing and calculating the salaries of Tagum Trio employees. The salary is based on a piece-rate system, where employees are paid according to their output. The production head is responsible for calculating each employee's salary in the production department using daily production records. The total salary is determined by multiplying the number of items produced by the corresponding rate of each product, which may vary depending on its width, size, and thickness.

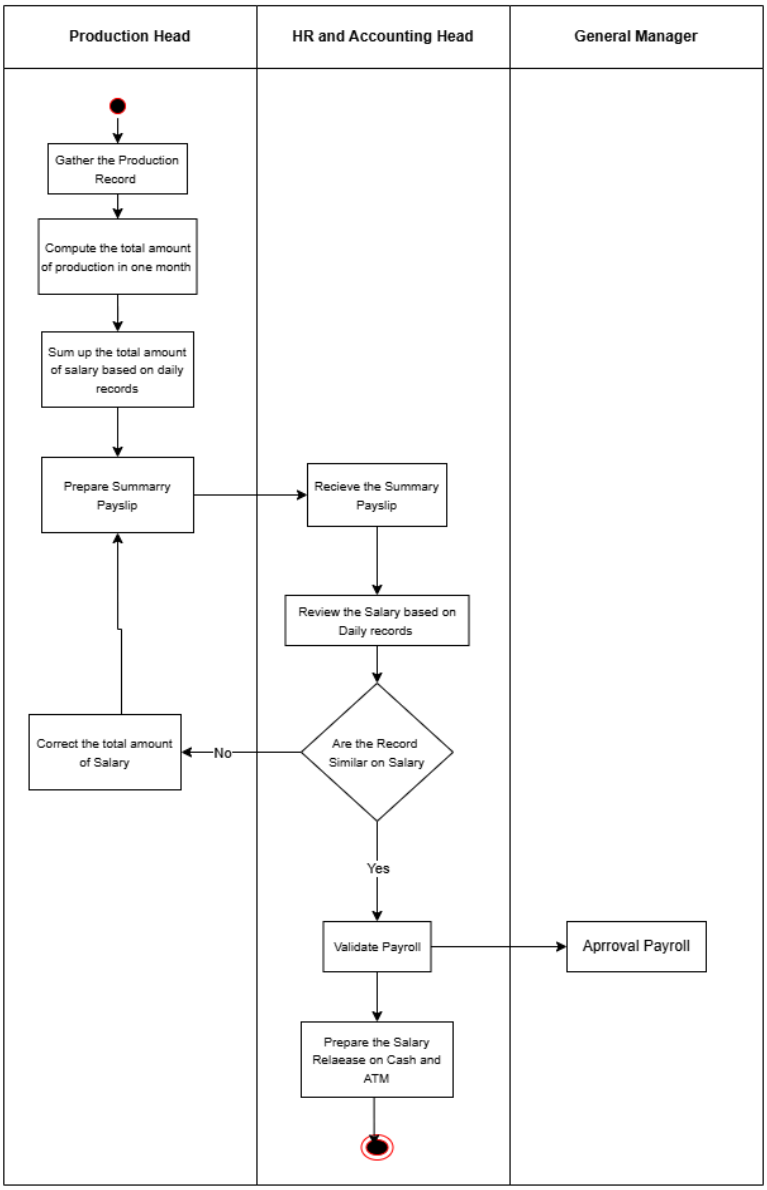

Figure 3. Payroll Processing

Transfer Department Processes

The provided figure illustrates the employee transfer process. , only the CEO and Manager possess the authority to transfer employees between production departments. These transfers are typically based on their observations; if a department is understaffed, employees may be reassigned to cover the shortfall. Currently, these transfers are conducted verbally, lacking proper documentation. While a leadman can

assign tasks to these temporarily transferred employees, their compensation is calculated based on their individual output within that department. The transferred employee then returns to their original department and position the following day.

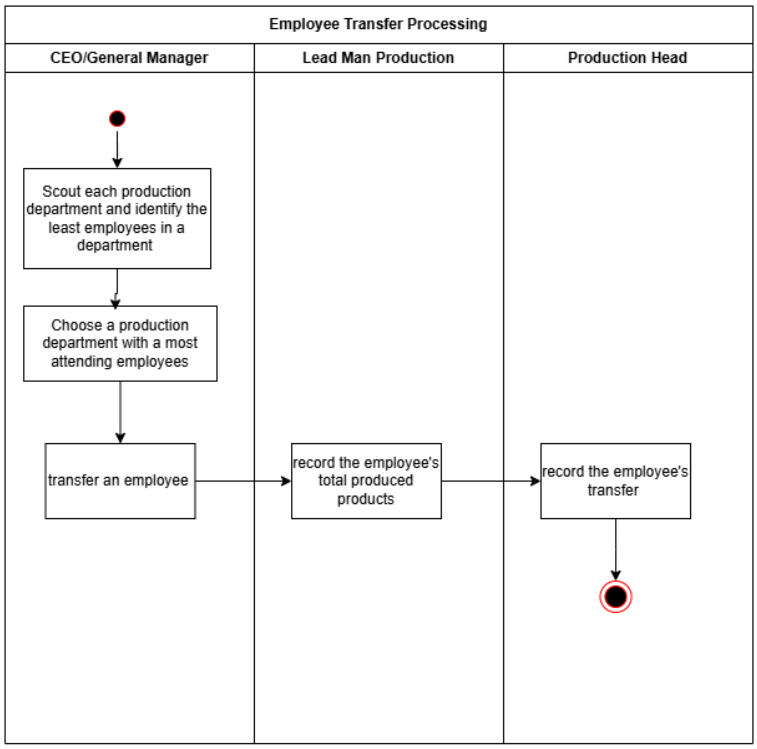

Figure . 4

Leave Processes

The illustration below details the company's leave approval process. A key step in this procedure, which typically extends the approval time to 2-3 days, involves the translation of the employee's leave request letter. This translation is essential to ensure that the General Manager or CEO, who may not be proficient in the original language of

the request, can fully understand the contents and make an informed decision regarding the leave application.

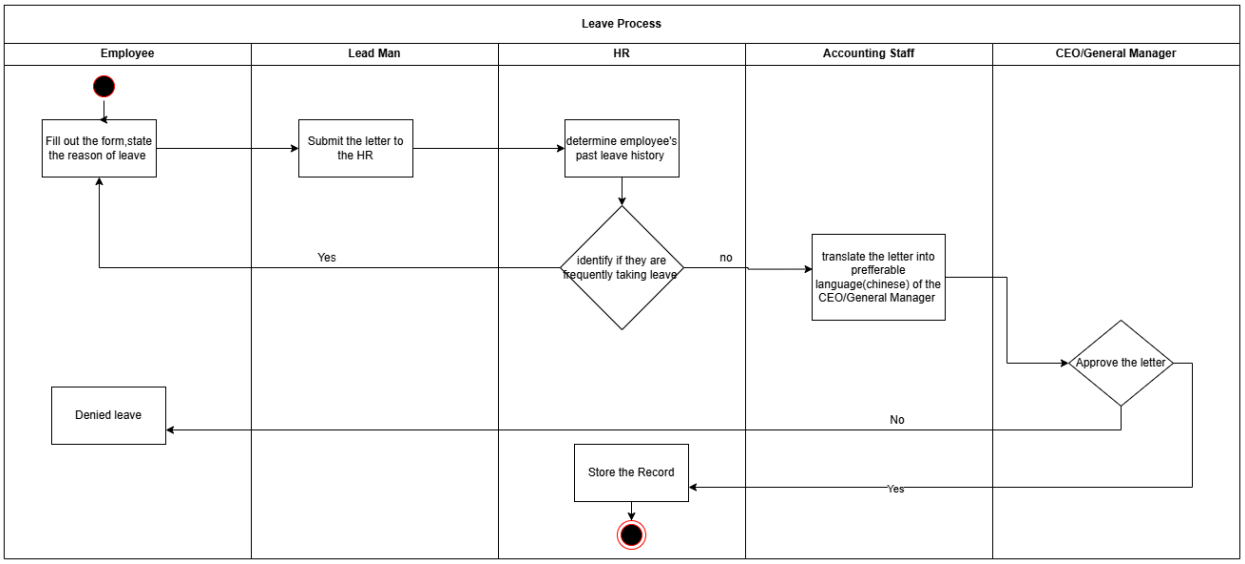

Figure 5. Current Consultancy Process

E. Objectives of the Project

a.General Objectives

The overall objective is to develop an integrated website system that centralizes data management, automates production recording and compensation calculations, streamlines operational processes, and facilitates clear communication and access controls, all to improve efficiency, accuracy, and transparency across the organization.

b.Specific Objectives

The following specific objectives aims to:

1. Implement an admin control module for secure login, credential management, user access rights, and system settings.
2. Develop a centralized employee information management module that securely stores, updates, and manages employee records, including personal details, work history, and supporting documents.

3. Establish a production oversight and management module for administrators to monitor real-time progress, identify operational concerns, and manage production sections.
4. Design a digital request management system that enables employees to submit various applications (e.g., leave, overtime, concerns) and allows administrators to review and approve employee compensation, work logs, and transfers.
5. Integrate a QR Code-based Production and Transfer Management System that allows leadmen to scan finished products for automatic record updates and calculation of employee salaries based on output, and enables the scanning of employees via QR code to officially log new transfers from different departments.
6. 7. Implement an employee communication and self-service portal for employees to access reminders, schedules, official announcements, and view their assigned sections and daily work details.
7. Provide a finance module for the finance department to review, approve, and generate reports for payslips.
8. Establish a comprehensive Workforce Analytics Platform that leverages integrated data from attendance, production, and employee transfers to generate insightful reports and dashboards, enabling management to optimize staffing, enhance productivity, identify training needs, and make data-driven decisions on workforce allocation and performance.

F. Scope and Limitation

G. Related Literature

Qr Production Process

In the old document management system, information was entered manually into the logbook, which wasted time and increased the likelihood of errors in recording information. This process is inefficient and unsuitable for larger companies or institutions with a significant number of documents. Since many employees tend to lose track of document paths, these systems should have the capability to trace the journey of documents from their source to their destination [8].QR codes, widely used for document tracking, are compact, square-shaped barcodes that can be scanned effortlessly with a smartphone or other QR scanner. This allows users to quickly find information or records related to the code. QR codes offer many benefits for document tracking, can involve expedient and effortless access to, as well as sharing of, information, and they easily integrate with existing records management systems [9].Modern information technology is changing the way we do business. They provide fast and seamless

communication between people, allowing them to exchange information anytime and anywhere. The introduction of the Internet, wireless networks, and mobile devices has contributed to the development of technologies that make our lives more convenient and productive. Digital cameras, smartphones, tablets, and other mobile devices have become an integral part of both our lives and production, allowing us to keep abreast of developments in science and technology [10-12]. QR codes in manufacturing play an important role in data management, product tracking and improving production processes.The widespread use of smartphones and other devices has led to an increase in the use of QR codes, so the topic of QR codes in science and technology is relevant.

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Salary Management

Every organisation keeps a record of their staff. Staff records play a crucial role in staff management. Every organisation requires these records to calculate pay, manage workforce and see performance of employees [13]. Management of all these records is a challenging task and time consuming process for the HR team [14] , which can be reduced by using EMS that is the Employee Management System [15]. Human resources are an important part of any organisation and responsible for the success of an organisation. Organisations invest highly on the management of employees. HRIS is a human resource information system that manages inventory control and accounting [16]. EMS is also an informatics system that is useful in saving time, power and money of owner, HR’s and manager. All organisations both private and public[15], have

necessities of employee management systems. But from years they have been using an old classical method that is pen and paper to maintain record, however in the past few years there has been a large increase toward using automatic systems[11v b] that can manage salary calculation by their own.

In the rapidly evolving landscape of public sector administration, the need for efficient, transparent, and secure information management systems is paramount. One of the most critical areas requiring improvement is the management of employee payroll, a process that directly impacts the satisfaction of government employees and the operational efficiency of government institutions. Traditional desktop-based systems have long been used for payroll management; however, they often suffer from issues such as slow processing times, error-prone manual data entry, limited accessibility, and lack of integration with other systems. As governments seek to modernize their operations, transitioning to web-based salary management systems has become an essential step in improving administrative functions (Suyanto, 2017; Aritonang, 2016).The implementation of web-based salary management systems can address these challenges by automating processes, enhancing real-time data access, and ensuring better security measures. A growing body of research highlights the significant advantages of adopting such systems, with several studies focusing on the use of advanced frameworks like the Zachman Framework in developing robust, scalable, and integrated solutions for public institutions (Kadir & Yusuf, 2021; Fitzgerald & Smith, 2021; Marshall & Daneshvar, 2020)

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[15] Sadhana J. Kamatkar , Amarapali Tayade , Amelec Viloria ,
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Workforce Analytics

Data has become a critical asset in the modern business landscape, driving decision-making processes across industries. Organizations increasingly rely on advanced analytical tools to gain insights that enhance efficiency, foster innovation, and sustain competitiveness (Gade, 2021) [16]. Among the most transformative tools in this context are workforce analytics and big data, each contributing uniquely to organizational success (Vassakis, Petrakis, Kopanakis, 2018) [32]. Workforce analytics, which focuses on leveraging data to optimize human resource management, provides organizations with the means to understand and address complex workforce challenges. It enables the strategic alignment of talent acquisition, performance management, and employee retention with broader business objectives, fostering a dynamic and adaptable workforce (Hosen et al., 2024) [18].At the same time, big data has fundamentally reshaped operations across sectors by enabling the collection, storage, and analysis of vast amounts of information. Big data allows organizations to uncover hidden patterns, forecast trends, and implement proactive strategies. For example, its applications in supply chain management have been particularly transformative (Vassakis et al., 2018) [32]. By leveraging data-driven insights, companies can optimize inventory management, improve demand forecasting, and enhance overall operational resilience. These capabilities are especially critical in complex supply chains such as those in the pharmaceutical industry, where precision and reliability are paramount (Saggi & Jain, 2018) [26].The convergence of workforce analytics and big data presents an opportunity for organizations to enhance decision-making capabilities. While workforce analytics focuses on optimizing human capital, big data extends its reach to broader operational domains, including supply chain management (Sharma & Khan, 2022) [27]. Together, these tools create synergies that enable organizations to address interdependent challenges, align strategies across functions, and build resilience in the face of dynamic market conditions (Rahaman & BARI, 2024) [24]. For instance, understanding workforce trends can directly impact supply chain efficiency by ensuring the availability of skilled personnel during critical

operational periods. Conversely, insights from supply chain data can inform workforce planning by identifying areas that require targeted interventions or skill development (Tiwari, Wee, & Daryanto, 2018) [30]

H. Related Systems

Qr Production Process

At Surigao del Norte State University, the process of keeping track of records is done manually. The current document tracking process involves offices: Office 1 receives incoming documents and transfers them to Office 2. The process relies on physical logbooks to keep track of each document's movement.Here is how it works: Office 1 records incoming documents in the logbook. When the documents are ready to be sent to Office 2, they are recorded again. Office 2 receives the documents, records them in their logbook, and processes them. After processing, Office 2 prepares the documents for transfer to another location, records them once more in the logbook, and obtains a signature. In the provisions of the Republic Act No. 11032 Section 9 [3], officers and receiving offices must fulfill their responsibilities in the efficient delivery of government services. The fault of the staff shall be impacted to the Office Heads as per stated in Section 8 [3].With this, there is a need to come up with a better way to keep track of records and information using technology. One idea is to create a system that centralizes all the records in one place, making it easier to manage and track them. This system would use QR codes to keep track of where the records are and their history, including where they came from, where they are now, and how long it takes for them to be processed. It would be a web-based application built using the PHP Laravel framework, capable of storing different types of documents and allowing them to be stored both locally and in the cloud.Records-tracking management systems are a type of information and communication technology (ICT) innovation that uses software to improve the way organizations manage and track their records. By providing a centralized platform for managing and tracking records, these systems can help organizations improve efficiency and transparency [4]The University of the Philippines (UP) employs a document tracking system (DTS) for enhanced efficiency and transparency [1]. The UP DTS lacks route functionality, causing potential delays [1]. Incorporating routes would significantly improve its effectiveness. QR codes,

increasingly popular due to their touchless nature, connect physical and digital realms [2]. Customers (75%) express a strong preference for continued QR code use [2].Document tracking can burden workers, causing displacement and missed deadlines [5][6]. Web-based systems offer a solution, aiding workplace efficiency [5]. Lingaya suggests a tracking model for Philippine education institutions, easing document management [6]. Google Drive, recommended by Muhammad, is a secure and widely-used option [7][8][9].

Salary Management

The salary management module. The financial department could use this module to maintain conveniently payroll records of the faculty, from five aspects, which are basic settings, payroll input, payroll audit, payment of salaries, and treatment at the end of a month. For basic settings, this module should provide functions of parameters modification, such as the income tax rate, and items modification, such as public reserve funds in gross wages, to the financial department, in the calculation formula of wages. For payroll input, this module should offer functions of Excel files uploading, batch operations, and manual inputs to the financial department. For payroll audit and payment of salaries, this module should supply specific users with corresponding operating permissions. For treatment at the end of a month, this module should consider how to conduct efficient and effective financial statements(1)Numerous studies have been conducted exploring various aspects of payroll systems and utilizing diverse techniques. Singh et al. [3] presented a study titled “automated payroll system (A-PAY),” aimed to digitize the payroll process for faculty members, including the management of their personnel information and the calculation of allowances and deductions. The study utilized Vb.net as the front end and Microsoft Access 2007 with SQL Server 2008 as the back end. The result was a more efficient system with options for data backup and restoration, improved security, reduced calculation errors, and automatic report generation. Hikmah and Muqorobin [4] developed an employee payroll information system for a Consulting Engineering Services company. They aimed to eliminate the challenges of manual payroll calculation. They conducted research through observation, interviews, documentation, and literature review, and used PHP and a MySQL database to build the system. The design improved payroll management and provided faster and easier retrieval of employee payroll information, leading to improved administrative

processing efficiency. In the study conducted by Lu et al. [5], a comprehensive approach to the design and implementation of a three-tier salary management system for higher vocational colleges was presented. The existing system was found to be complex and time-consuming, leading the researchers to aim for a more efficient and accurate computerized payroll system. The research process involved analyzing the current status, needs, and requirements, and implementing the design using ASP.NET. The results indicated that the study's objectives were successfully achieved, as the developed system was efficient, time-saving, and enabled the organization to manage payroll effectively. Winanda et al. [6] presented a study titled "salary information system." The authors identified inefficiencies in the manual salary calculation system at Tiara Handicraft Enterprise, including high labor requirements and inadequate communication between departments, leading to limited data validity. The objective of their research was to design and implement a new system that addressed these issues. The results indicated that the new automated system was capable of more efficiently and effectively calculating salary data.

Workforce Analytics

According to Huselid (2018), workforce analytics shifts human resource management from intuition-based decision-making toward evidence-based practices. Its international relevance is grounded in global industrial challenges such as labor shortages, high turnover, and fluctuating productivity levels in manufacturing sectors across both developed and emerging economies. The systematic deployment of workforce analytics is facilitated by Management Information Systems (MIS), which serve as digital platforms to collect, store, and interpret human capital data (Levenson, 2017). MIS applications offer operational dashboards, real-time tracking, and predictive reporting, which enable managers to align workforce capabilities with production requirements. Particularly in the manufacturing domain, where operational efficiency is directly linked to labor input, workforce analytics contributes significantly to overall equipment effectiveness (OEE), labor cost control, and compliance (Huselid et al., 2025).In manufacturing environments, labor planning is a key function of workforce analytics that aims to match human resource capacity with production schedules while minimizing costs and disruptions (Clark, 2017). Labor planning encompasses a range of activities including shift

scheduling, overtime management, workforce allocation, and demand forecasting. Management Information Systems facilitate this process through tools such as Human Resource Information Systems (HRIS), Enterprise Resource Planning (ERP) modules, and workforce scheduling algorithms (Omar et al., 2019). For instance, the use of real-time labor dashboards integrated into ERP systems allows plant managers to visualize workforce capacity in relation to production line requirements. Moreover, simulation-based labor planning models, supported by MIS, enable scenario analysis to predict labor needs under varying production loads (Cao & Duan, 2017). Empirical evidence from global manufacturing firms shows that integrating workforce analytics into labor planning enhances workforce availability and reduces operational downtime. The international application of such tools is evident in lean manufacturing environments across Germany, Japan, and the United States, where labor cost management is critical to competitive advantage. In these contexts, labor planning systems not only support scheduling but also incorporate worker qualifications, shift preferences, and fatigue levels, thereby aligning human factors with productivity goals (Porter & Heppelmann, 2014).

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CHAPTER II

METHODOLOGY

A. Project Initiation

a. Software Development Life Cycle

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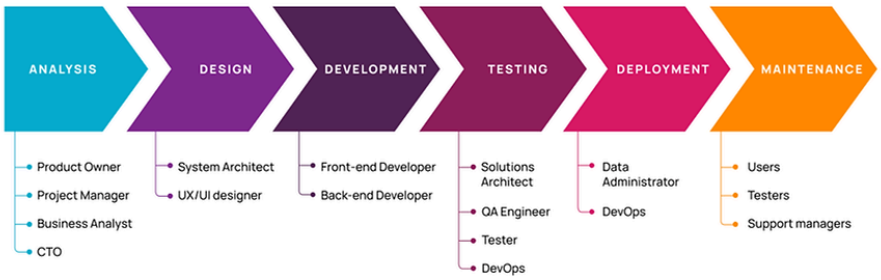

Figure X. SDLC Phases Sample

b. Work plan and Schedule Feasibility

Project Milestones and Deliverables

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Integrated and Proposed New Processes

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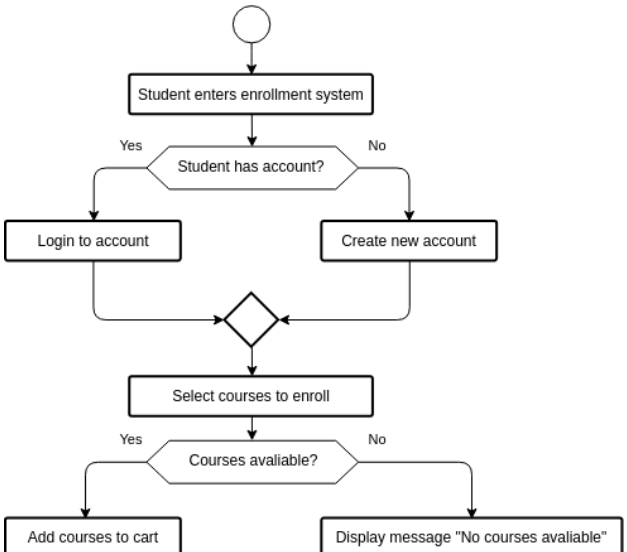

Figure X. Current Consultancy Process Sample

(Add more process if needed)

Development Team

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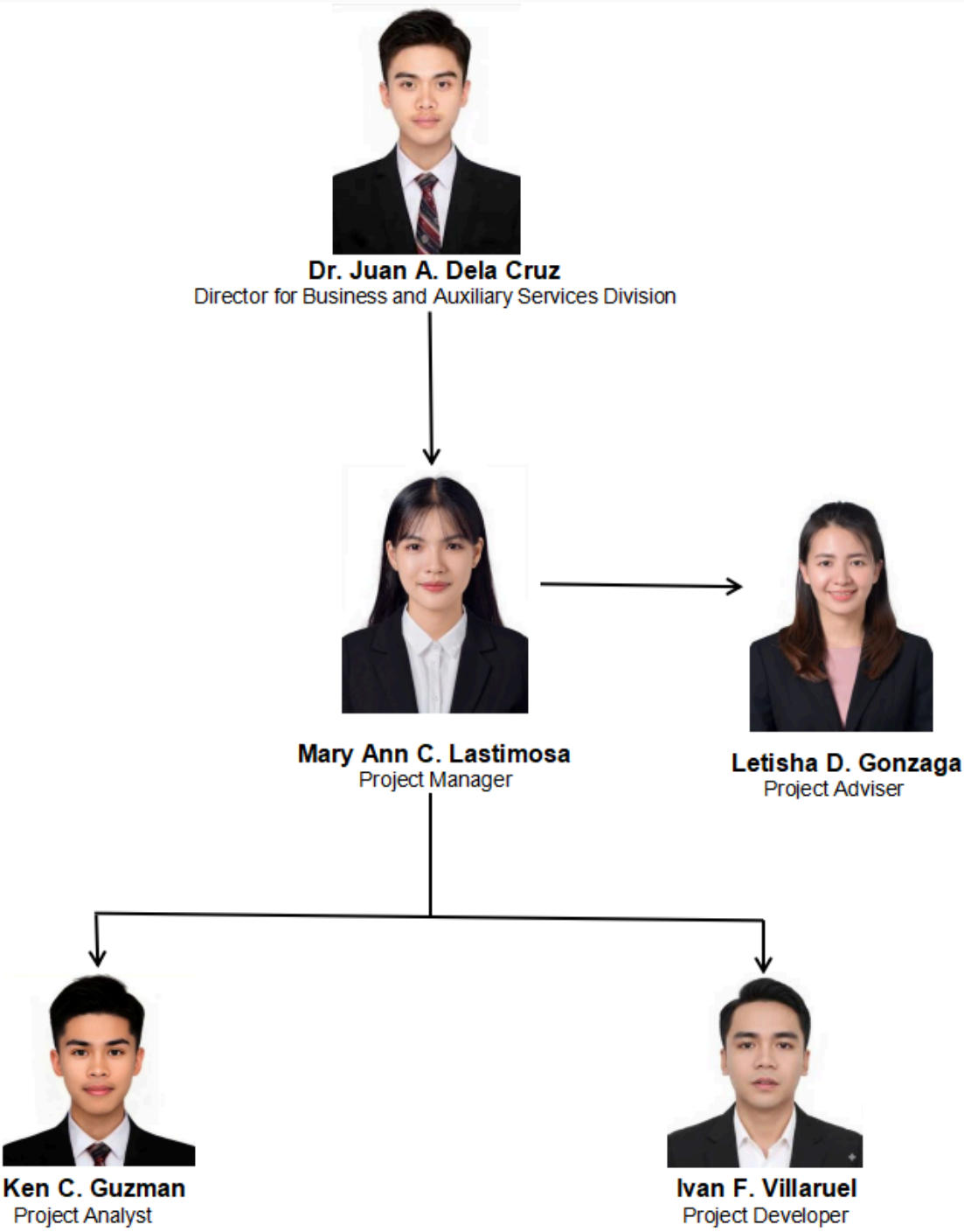

Figure X. Project Team Organizational Structure

(Disclaimer: The picture provided is for sample purposes only and should not be distributed.)

c. Feasibility

Operational Feasibility

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Technical Feasibility

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massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

Table X. Business's Existing Technology Template

Technology	Particulars and Specification	Quantity
Software		
Hardware		
Network		

d. Stakeholder Analysis

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Table X. Stakeholder Analysis Template

e. Project Charter

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Table X. Project Template

f. Project Team Roles and Responsibilities

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massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

Table X. Roles and Responsibility Template

g. Work Breakdown Structure

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a. Work Breakdown Structure

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Figure X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

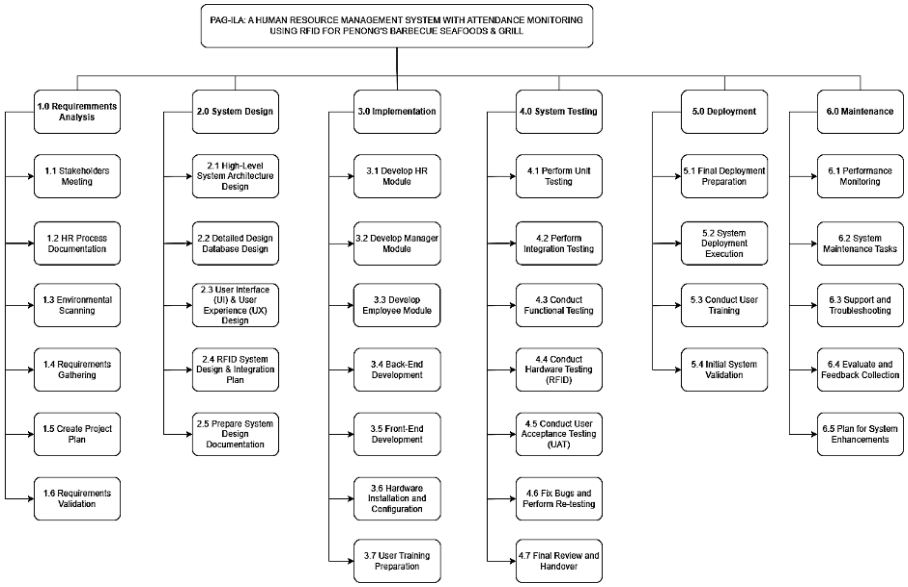

Figure x. WBS Sample

b. **WBS Dictionary**

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Table X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

Table X. Work Breakdown Dictionary Template

WBS Code	Phases and Activity	Duration (Mandays)	Star Date	End Date	Actual Cost

h. Gantt Chart

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Figure X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

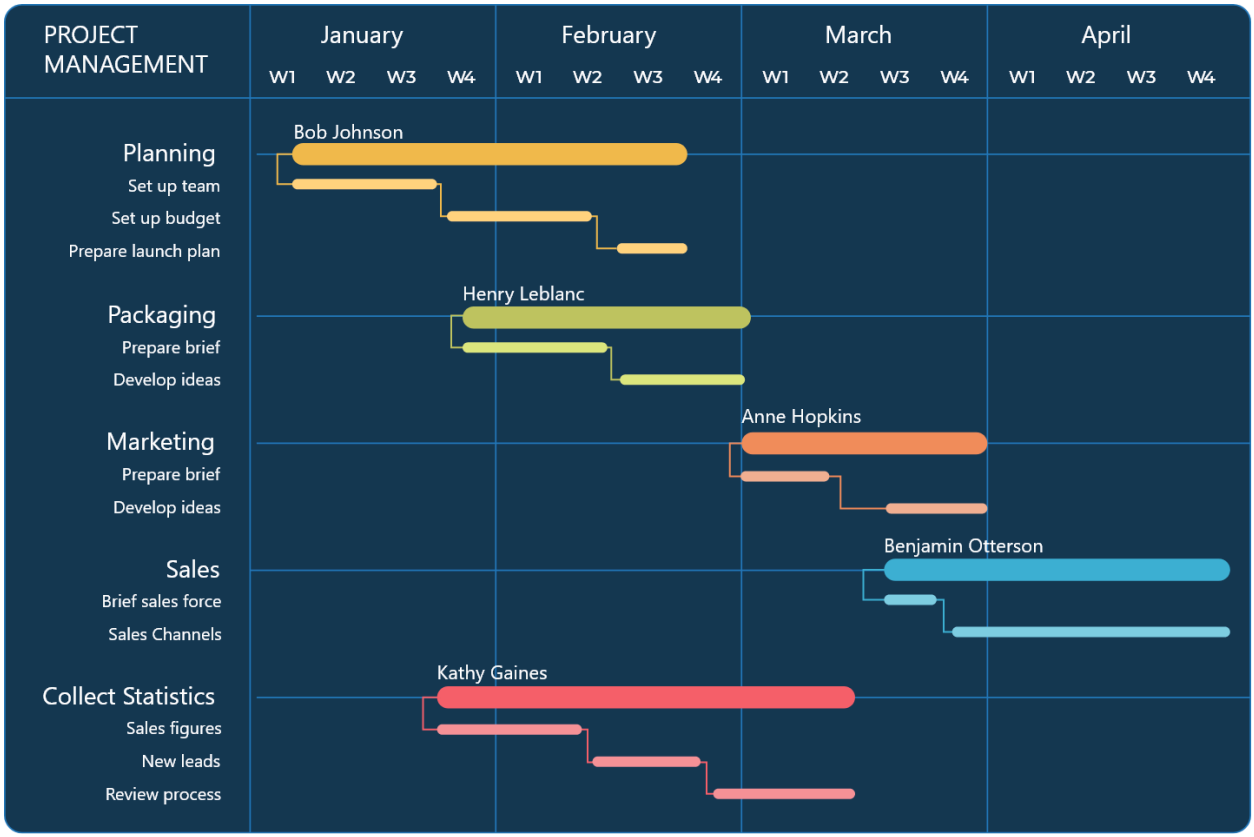

Figure X. Gantt Chart Sample

B. System and Requirement Analysis

a. Budget Summary

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Table X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

Table X. Budget Summary of the Capstone Project

Phase	Duration (Man-days)	Start Date	End Date	Actual Cost
Total Man-days		**Total Actual Cost**		

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Table X. Manpower Cost Estimates

Position	Man-days	Rate/Day	Actual Cost
Project Manager			
Project Developer			
Project Analyst			
		TOTAL	

b. Budget Allocation and Schedule

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Table X. Budget Allocation per Month

YEAR	MONTH	WBS COST	Technology Cost	Audit Cost	Effort Cost	Total Cost	Cumulative Cost
2026	February						
	March						
	April						
	May						
	June						
	July						
	August						
	September						
	October						
	November						
	December						
2027	January						
	February						
	March						
	May						
Total							

Effort Cost

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Table 10. Effort Cost per Resource

Role	Communication	Meeting	Reporting	Actual Cost
Project Manager				
Project Developer				
Project Analyst				
Total				

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Cash Outlay

Table X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

Table X. Hardware: Actual Cost

Hardware	Quantity	Unit Price	Total Cost
TOTAL			

Table X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

Table X. Software: Actual Cost

Software	Quantity	Actual Cost
	Total Cost	

c. Financial Viability

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d. Cost and Benefit Description

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Table X. Time Spent: Current vs. Proposed Process

Processes	Current	Proposed	Difference
Total			

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Table X. Expenses: Current Process of Facility Manager

Processes	**Facility Manager (**₱76.25/hour)	TOTAL
Total		

(Add more tables based on your users)

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Table X. Facility Booking: Current vs. Proposed

Facility Booking	Cost	Average Customer	Cost per Week	Cost per Month	Cost per Year
Current	₱76.25	10	₱5,337.50	₱21,350.00	₱256,200.00
Proposed	₱12.71	10	₱889.70	₱3,558.80	₱42,705.60
Difference	**₱63.25**			**₱17,791.20**	**₱213,494.40**

(Add more tables based on your process)

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Table X. Cumulative Cost VS Cumulative Savings Until Break-Even

YEAR	MONTH	CUMULATIVE COST	SAVINGS	CUMULATIVE SAVINGS

e. RACI Chart

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Table X. RACI TABLE

Legends	
Responsible	Assigned to complete the given task
Accountable	Has the final decision-making and accountability. Only one person per task.
Consulted	An adviser and shareholder in decision-making.
Informed	Must be informed about the decision and action.

Rental Management System for Davao del Norte State College (DNSC)						
Main Tasks and Milestones	**Project Manager**	**System Analyst**	**System Programmer**	**System Designer**	**User Tester**	**Project Sponsor**

f. System Analysis

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System Requirements (Functional and Nonfunctional)

Functional Requirements:

- Lorem ipsum dolor sit amet, consectetur adipiscing elit.
- Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
- Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
- Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

Non-Functional Requirements:

- Lorem ipsum dolor sit amet, consectetur adipiscing elit.
- Nulla facilisi. Ut faucibus egestas interdum.
- Integer nec odio. Praesent libero. Sed cursus ante dapibus diam.

- Aenean massa. Nulla metus metus, ullamcorper vel, tincidunt sed, euismod in, nibh.

Process Model: Use Case Diagram and Description

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Figure X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

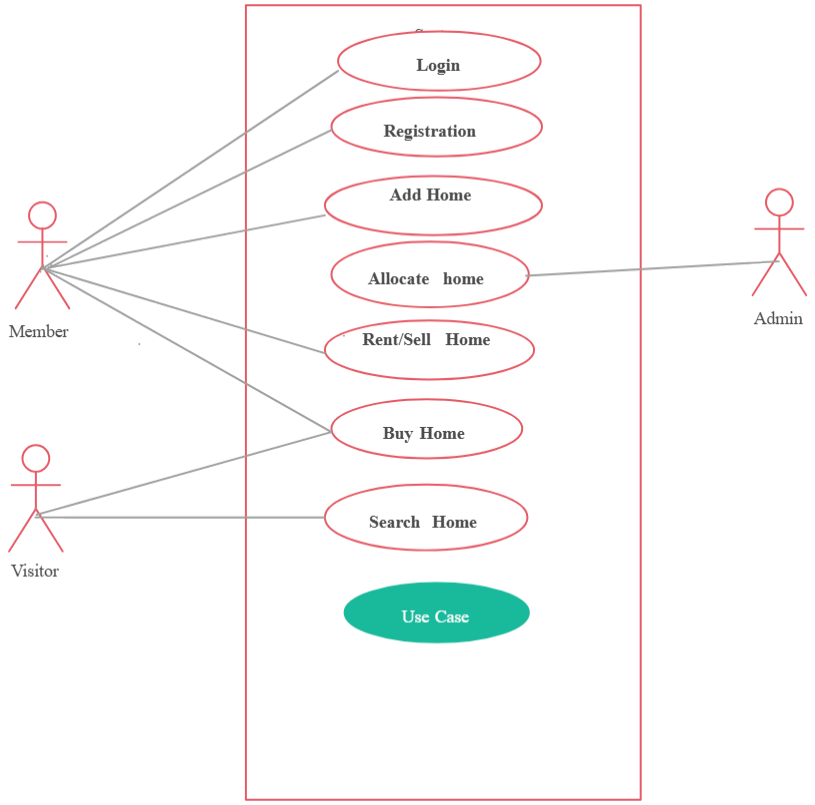

Figure X. Use Case Diagram

Table X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

Table X. Use Case Description for Registration

Use Case Name	Registration
Participating Actors	● Member
Entry Conditions	The user has no existing account or intends to create a new one.
Flow of Events	1. The user accesses the registration form. 2. The user enters required personal details such as Full Name, Email Address, and Password. 3. The user agrees to the terms and conditions by checking the acceptance box. 4. The user submits the registration form. 5. The system validates the input data (e.g., checking for email format and password strength). 6. If the data is valid, the system stores the user information and creates an account. 7. The system sends a confirmation email to the user for account verification. 8. The user clicks the verification link in the email to confirm the registration. 9. The system activates the user account, allowing them to log in..
Exit Conditions	The user receives a confirmation email to verify their account.
Quality Requirements	The system must ensure data security by using encryption for sensitive information such as passwords.

(Add more tables based on your use case name)

Process Model: Activity Diagram

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(Insert Name of the Process here)

Figure X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

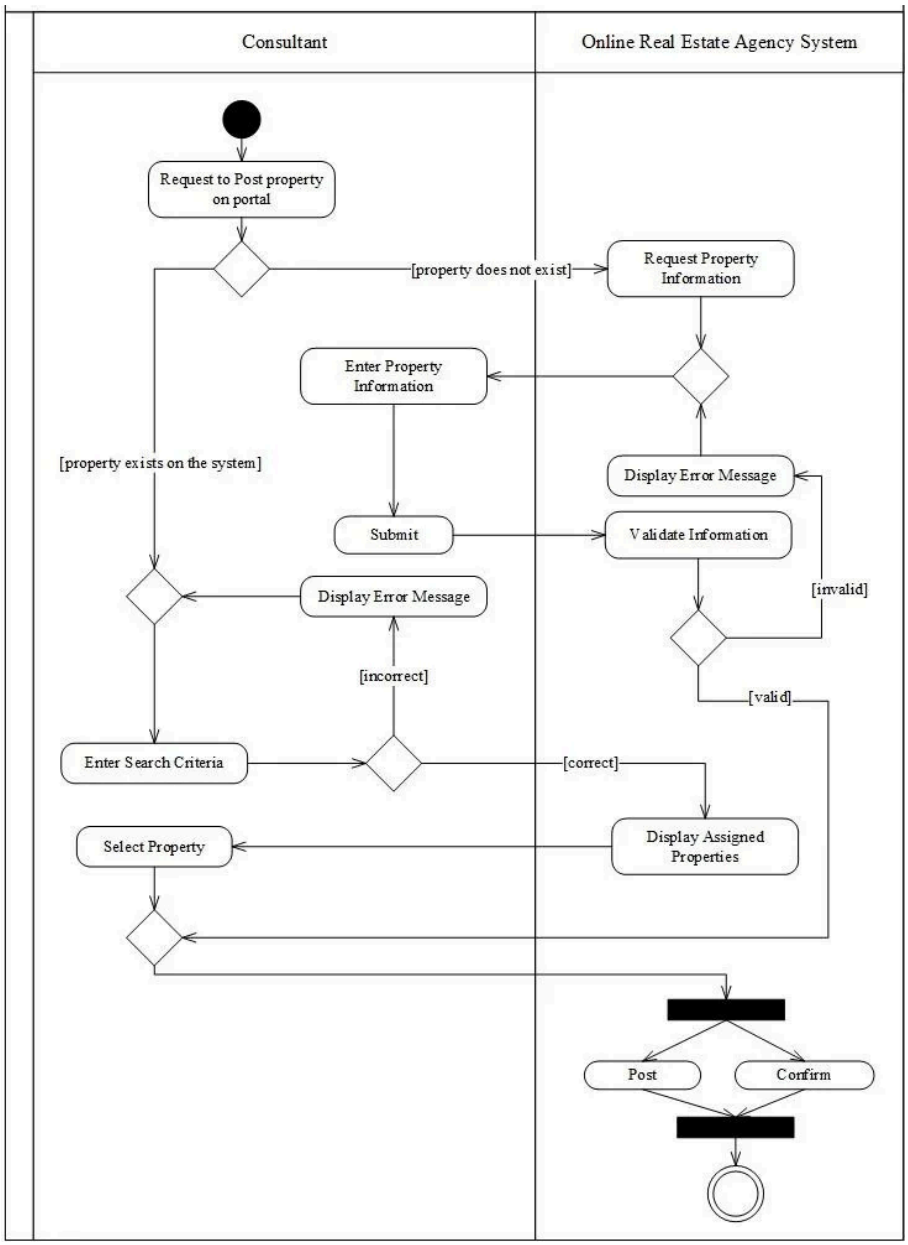

Figure X. Online Process Activity Diagram

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

(Add more diagrams based on the process)

Data Model: Context Data Flow Diagram

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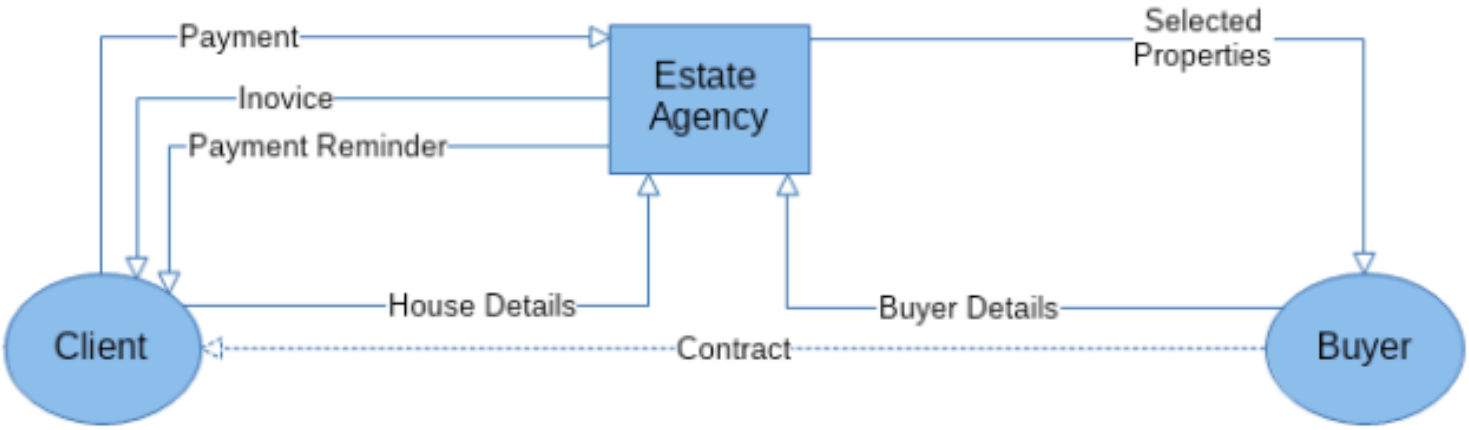

Figure X. Context Data Flow Diagram

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

Data Model: Level 1 Data Flow Diagram

The Level 1 data flow diagram for the system is Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

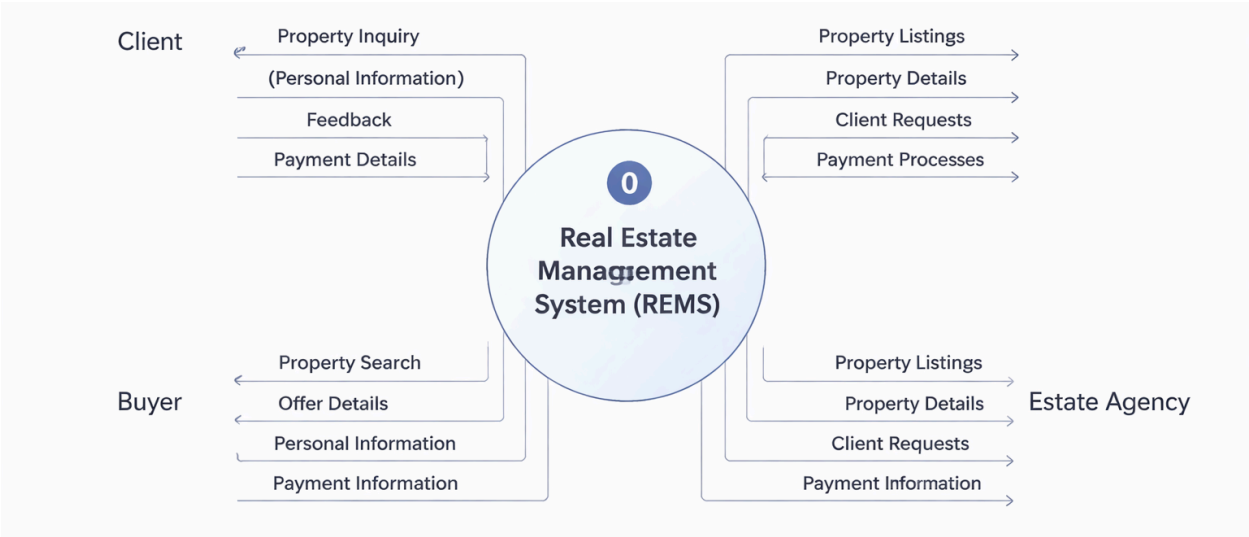

Figure X. Level 1 Data Flow Diagram

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

C. System Design

a. Database Design: Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

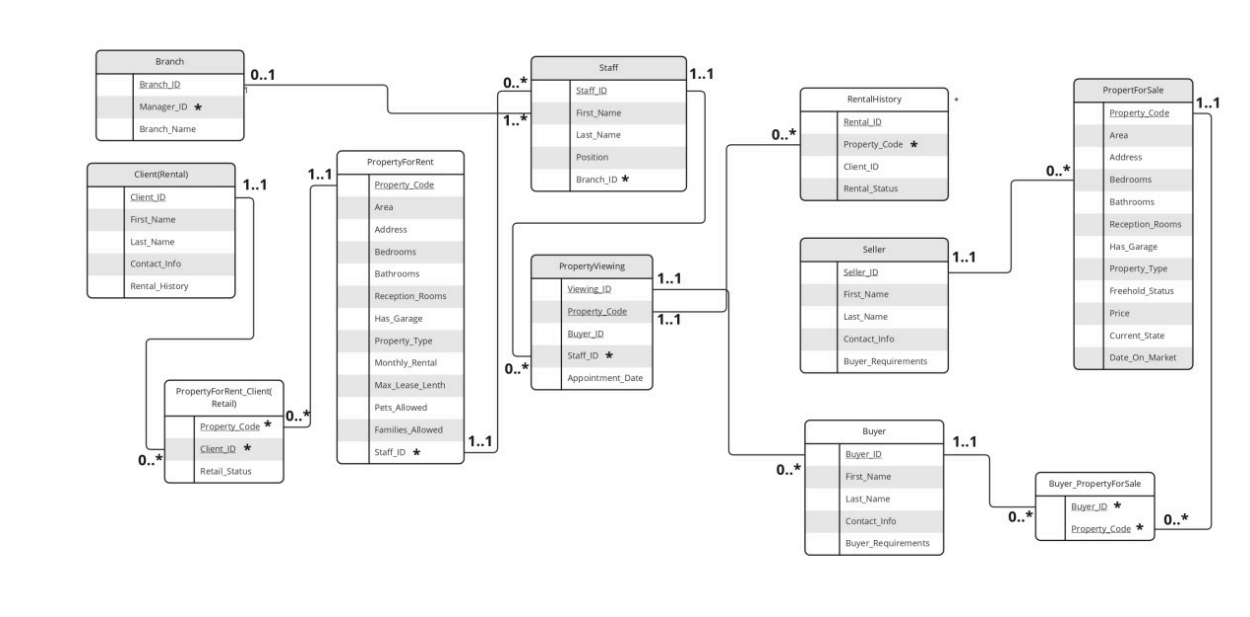

Figure X.Entity-Relationship Diagram

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

b. Data Dictionary

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Table X. Data Dictionary

Field Name	Data Type	Field Length	Constraint	Description
booking_id	int	11	PRIMARY KEY	Unique ID for the booking
start_date	date		NOT NULL	Booking's check-in date
end_date	date		NOT NULL	Booking's check-out date
start_time	time		NOT NULL	Booking's check-in time
end_time	time		NOT NULL	Booking's check-out time
school_year	varchar	20	NULL	Academic year
semester	varchar	10	NULL	Academic semester
organization	varchar	255	NULL	Organization making the booking
contact_info	varchar	255	NULL	Contact information
purpose	varchar	255	NULL	Purpose of the booking
num_users	int	11	NULL	Number of users
equipment_needed	text		NOT NULL	Equipment needed for the booking
activity_design	Varchar	255	NOT NULL	Activity plan requirement
request_letter	varchar	255	NOT NULL	Request Letter contents
status	varchar	20	NULL	Status of the booking
created_at	timestamp		NOT NULL	Record was created

(Add more tables based on your entity mentioned in the ERD)

c. System Architecture

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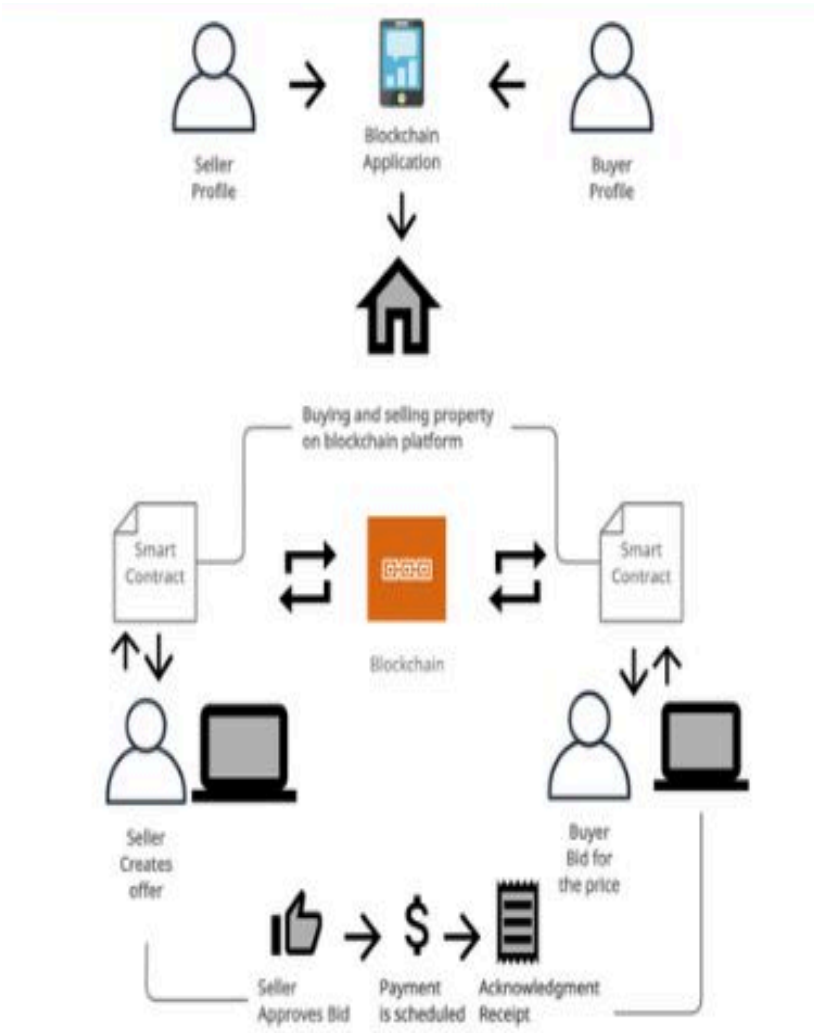

Figure X. System Architecture

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

D. System development and deployment

a. Test Cases, Alpha Testing, and UAT

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b. Deployment Plan with Sampling Technique

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c. Questionnaire

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E. System Maintenance

a. Risk Management Plan

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Table X. Risk Management Plan for the (Name of your system)

Risk ID	Risk	Description	Likelihood	Severity	Risk Factor	Actions to Minimize Risk
1	**Internal Risks**					
1.1						
1.2						
1.3						
1.4						
1.5						
2	**External Risks**					
2.1						
2.2						
2.3						
2.4						
2.5						

b. Project Turnover Plan

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Table X. Project Closure Report

PROJECT CLOSURE REPORT				
Project Name				
Brief Description				
Project Sponsor				
Prepared by	Project Team			
Objectives Achievement				
Objectives	**Description**	**Performance**	**Achieved**	**Not Achieved**
ONLINE FACILITY BOOKING				
INCOME AND FACILITY OCCUPANCY REPORT				

Approvals
This document requires the approval:

Position	Signature	Date
Project Manager		
Project Sponsor		